

tained a pressure sensor, a transistor and a tiny LED. Pressure changed the resistance of the sensor, thereby changing the amount of electricity flowing into the LED.

The team suggested that the invention could be useful in skin for prosthetic limbs and robotics. One of the major problems with these kinds of light films in the past, though, was that these only lasted a matter of hours when exposed to normal air.

Making any piece of ultra-stretchy electronics often involves sandwiching materials together to produce something with the right properties, whether it's for a red light or a method of sensing pressure. In this case, the researchers added a new protective coating, called a passivation layer, to various kinds of e-skin. The coating kept out oxygen and water vapour well enough to keep the light working for several days. Researchers report that this power LED film also produced less heat and consumed less power than previous efforts. The coating they used can also work on e-skin that does more than just light up.

Sweating electronic skin

A Northwestern University research team has developed a first-of-its-kind soft, flexible microfluidic device that easily adheres to the skin and measures the wearer's sweat to show how his body is responding to exercise.

A little larger than a quarter and about the same thickness, the simple, low-cost device analyses key biomarkers to help the user decide quickly if any adjustments, such as drinking more water or replenishing electrolytes, need to be made or if something is medically awry.

Designed for one-time use of a few hours, the device, placed directly on the skin of the forearm or back, even detects the presence of a biomarker for cystic fibrosis. In the future, it may be more broadly used for disease diagnosis.

Sensory electronic skin

Researchers at the Ulsan National Institute of Science and Technology have developed an electronic skin that can detect changes in both tem-



Electronic skin that lights up when touched



Northwestern University's flexible device measures the wearer's sweat to analyse his key biomarkers



Leibniz Institute's skin provides a sense of magnetic fields

perature and pressure. The skin's composites are made of a polymer as well as reduced graphene oxide. Thanks to that, the artificial skin can detect not only sensory changes created by water droplets falling at different pressures and temperatures but also the tiniest amount of pressure such as human hair's. This makes the skin perfect for future caregiver robots. Or, when attached to a human wrist, it can also be used to monitor pulse pressure by detecting the changes in skin temperature that occur when blood vessels dilate or constrict.

Researchers have created a wear-

able device that is as thin as a temporary tattoo and can store and transmit data about the wearer's movements, receive diagnostic information and release drugs into skin. Similar efforts to develop electronic skin abound, but the device is the first that can store information and also deliver medicine—combining patient treatment and monitoring. Its creators say that the technology could one day aid patients with movement disorders such as Parkinson's disease or epilepsy.

The researchers constructed the device by layering a package of stretchable nanomaterials—sensors that detect temperature and motion, resistive RAM for data storage, micro-heaters and drugs—onto a material that mimics the softness and flexibility of the skin.

Sensing magnetic fields

There are actually several more senses than humans' basic senses of touch, sight, hearing, taste and smell. One that humans don't inherently have is magnetoreception—the ability to perceive magnetic fields. Found in certain bacteria, migratory birds, fish and invertebrates, it provides them a better sense of navigation and orientation.

Researchers at the Leibniz Institute for Solid State and Materials Research in Germany describe a new electronic skin that provides a sense of magnetic fields. The electronic skin contains an array of magnetoresistive sensor foils that sense both static and dynamic magnetic fields. The sensors are made from layers of cobalt and copper, with polyethylene terephthalate (PET) film. Information about the sensor's proximity to a magnetic field is transmitted wirelessly to an external device that has LED indicators, giving a visual representation of the distance.

The skin is only about two micrometres thick, which is about one-fifth as wide as a single human hair. A square metre of the material weighs only three grams, which makes it light enough to rest on a soap bubble. It is also incredibly elastic, as it is able to stretch over 270 per cent in multiple directions over 1000